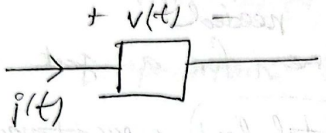
Topic → Power Calculation



$$p(t) = v(t)\, i(t) = \frac{\text{Energy}}{\text{time}}$$

$$\Rightarrow E = pt = \int_0^T p\,dt = \boxed{\int_0^T v(t)\, i(t)\, dt}$$

$$\therefore P = \frac{E}{t} = \frac{E}{T} = \boxed{\frac{1}{T}\int_0^T v(t)\, i(t)\, dt}$$

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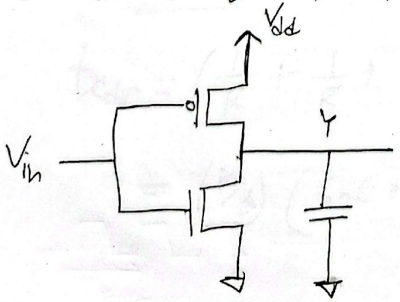
calculate power for an inverter that is driving a capacitor as its load:



when V_{in} 1 হতে 0 করলে → cap. charges from 0 to V_{DD}. Fully V_{DD} তে charged হলে আর কোনো current flow হতে না as দুমাথার potential same হয়ে গিয়েছে।

So, 1 হতে 0 করলে → charge হওয়া time এ কিছু current flow করে for some time.

এখন V_{in} = 0 হতে 1 করলে → Capacitor starts discharging hence I flows from cap. to NMOS to GND. After fully discharging, I flow stops at some moment.

এইভাবে capacitor charge নিয়ে রাখে তার energy store রাখার concept এখান থেকেই পাই।

$$E_{cap} = \int v(t)\, i(t)\, dt$$

$$= \int v(t) \left\{ C \frac{dv}{\cancel{dt}} \right\} \cancel{dt}$$

$$= C \int_{v=0}^{V_{DD}} v(t)\, dv$$ → during charging

$$= C \cdot \left[\frac{v^2}{2}\right]_0^{V_{DD}} = \left(\frac{1}{2} C V_{DD}^2\right)$$

→ charging time এ এই $\frac{1}{2}CV_{DD}^2$ energy consumed হচ্ছে by the capacitor. (or stored)

→ Source তথা power supply (V_{DD} here) কর্তৃক energy supply করছে?

$$\Rightarrow E_{V_{DD}} = \int V(t)i(t)\,dt = \int_0^{T/2} V_{DD} \cdot C\frac{dV}{\cancel{dt}}\,\cancel{dt}$$

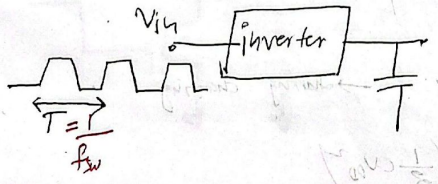
$$= CV_{DD} \int_0^{V_{DD}} dV$$

$$= (CV_{DD}^2)$$

তাহলে বাকি $(\frac{1}{2}CV_{DD}^2)$ কোথায় যাচ্ছে?

→ PMOS এ loss হয়ে গেটা।

$\text{Power} = \left(\frac{CV_{DD}^2}{T}\right)$ তাহলে প্রতিবার switching এর সময়ে total source এর power consumption হচ্ছে।



প্রতি T সময় অন্তর পর পর V_{in} 1 হলে 0 hence capacitor 0 হলে 1 হচ্ছে তখন $\frac{CV_{DD}^2}{T}$ power হচ্ছে।

$$\boxed{\text{Power} = (CV_{DD}^2) \times f_{sw}}$$ → freq of the input pulse switching of $V_{in}(t)$.

$\alpha = \left(\frac{f_{sw}}{f_i}\right)$ = activity factor বলে।

→ freq of system's clk freq

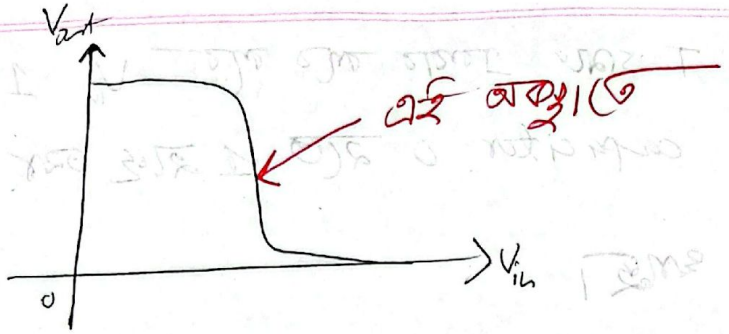
system এর clk freq এ কাজ করলে (f), inverter এও same freq (f_{sw}) এ on/off করলে, $\alpha = 1$ otherwise α কম হবে।

$$\boxed{P_{sw} = C V_{DD}^2 (\alpha f)}$$ → switching power

$P_{dynamic}$ = inverter যতগুলো used হবে ততগুলো consumed power

$= P_{sw} + P_{short\,ckt}$

in inverter's VTC curve, when both NMOS, PMOS are in sat mode, a short ckt occurs from V_{DD} to GND.



P_{static} = when inverter is totally off (input এ কিছু নাই)

= (occurs even when $V_{gs} < V_{tn}$ or $V_{sg} < |V_{tp}|$ due to subthreshold leakage current at gate)

+ (thin gate oxide এর কারণে কিছু charge leak করে to the channel) + (Drain বা source হতে body তে যেতে পারে কিছু charge) + $P_{contention}$

$= P_{sub} + P_{gate} + P_{junct} + P_{contention}$

Example 5.1

A digital system-on-chip in a 1 V 65 nm process (with 50 nm drawn channel lengths and λ = 25 nm) has 1 billion transistors, of which 50 million are in logic gates and the remainder in memory arrays. The average logic transistor width is 12 λ and the average memory transistor width is 4 λ. The memory arrays are divided into banks and only the necessary bank is activated so the memory activity factor is 0.02. The static CMOS logic gates have an average activity factor of 0.1. Assume each transistor contributes 1 fF/μm of gate capacitance and 0.8 fF/μm of diffusion capacitance. Neglect wire capacitance for now (though it could account for a large fraction of total power). Estimate the switching power when operating at 1 GHz.

SOLUTION: There are (50×10^6 logic transistors)(12 λ)(0.025 μm/λ)((1 + 0.8) fF/μm) = 27 nF of logic transistors and (950×10^6 memory transistors)(4 λ)(0.025 μm/λ)((1 + 0.8) fF/μm) = 171 nF of memory transistors. The switching power consumption is $[(0.1)(27 \times 10^{-9}) + (0.02)(171 \times 10^{-9})](1.0\text{ V})^2(10^9\text{ Hz}) = 6.1$ W.

system clock freq, $f = (1 \times 10^9)$ Hz (given)

total transistors $\rightarrow (1 \times 10^9)$

logic gates $\rightarrow 50 \times 10^6$

$\therefore$ memory arrays $= (1 \times 10^9 - 50 \times 10^6) = (950 \times 10^6)$

(a) For logic gates transistors:

$\alpha_T = (0.1)$ given

we know, capacitance = width multiplier $\times C$

$\therefore$ total capacitance for logic transistors

$= (50 \times 10^6) \times (12\lambda) \times (1 + 0.8)\frac{fF}{\mu m}$ per width

$= 50 \times 10^6 \times (12 \times 0.025\,\mu m) \times (1 + 0.8)\frac{fF}{\mu m}$

$= (27\,nF)$

(b) for memory arrays:

total capacitance for memory transistors

$= (950 \times 10^6) \times (4 \times 0.025\,\mu m) \times (1 + 0.8) \times \frac{10^{-15}\,F}{\mu m}$

$= 171\,nF$

activity factor, $\alpha_M = (0.02)$ given

Hence, total switching power $= (\alpha_T f) C_T V_{DD}^2 + (\alpha_M f) C_M V_{DD}^2$

$= 0.1 \times 10^9 \times (27 \times 10^{-9}) \times (1)^2 + 0.02 \times 10^9 \times (171 \times 10^{-9}) \times (1)^2$

$= (6.12)\,W$

Example 5.4

Consider the system-on-chip from Example 5.1. Subthreshold leakage for OFF devices is 100 nA/μm for low-threshold devices and 10 nA/μm for high-threshold devices. Gate leakage is 5 nA/μm. Junction leakage is negligible. Memories use low-leakage devices everywhere. Logic uses low-leakage devices in all but 5% of the paths that are most critical for performance. Estimate the static power consumption.

SOLUTION: There are $(50 \times 10^6$ logic transistors$)(0.05)(12\ \lambda)(0.025\ \mu\text{m}/\lambda) = 0.75 \times 10^6$ μm of low-threshold devices and $[(50 \times 10^6$ logic transistors$)(0.95)(12\ \lambda) + (950 \times 10^6$ memory transistors$)(4\ \lambda)](0.025\ \mu\text{m}/\lambda) = 109.25 \times 10^6\ \mu$m of high-threshold devices. Neglecting the benefits of series stacks, half the transistors are OFF and contribute subthreshold leakage. Half the transistors are ON and contribute gate leakage. $I_{sub} = [(0.75 \times 10^6\ \mu\text{m})(100\ \text{nA}/\mu\text{m}) + (109.25 \times 10^6\ \mu\text{m})(10\ \text{nA}/\mu\text{m})]/2 = 584$ mA. $I_{gate} = ((0.75 + 109.25) \times 10^6\ \mu\text{m})(5\ \text{nA}/\mu\text{m})/2 = 275$ mA. $P_{static} = (584\ \text{mA} + 275\ \text{mA})(1\ \text{V}) = 859$ mW. This is 15% of the switching power and is enough to deplete the battery of a hand-held device rapidly.

5.4

$I_{sub,low} = \frac{100\,nA}{\mu m}$, $I_{sub,high} = \frac{10\,nA}{\mu m}$

$I_{gate} = 5\,nA/\mu m$ for both low and high threshold devices.

$I_{junc} = 0$, $I_{contention} = 0$.

total logic transistors = 50×10^6

" memory " = 950×10^6

~~For memory transistors:~~

5% of logic transistors use high leakage ie low threshold devices. Total length occupied by such low threshold devices is $= \left(\frac{5}{100}\right) \times (50 \times 10^6) \times 12 \times 0.025\,\mu m = (0.76 \times 10^6)\,\mu m$

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rest 95% logic transistors and all memory transistors use low leakage ie high threshold voltage devices and total length occupied by such high threshold devices ~~$= (950 \times 10^6 + 95\% \text{ of } 50 \times 10^6)$~~

$= \{95\% \text{ of } 50 \times 10^6 \times 12 \underbrace{\lambda}_{=0.025\mu m}\} + \{950 \times 10^6 \times 4 \underbrace{\lambda}_{0.025\mu m}\} = (109.25 \times 10^6)\,\mu m$

let, half the transistors are ON and contribute to gate leakage. $P_{gate} = (I_{gate} \times V_{DD})$

$\therefore P_{gate} = \frac{1}{2} \times \left\{ 0.75 \times 10^6\,\mu m \times \frac{5\,nA}{\mu m} + 109.25 \times 10^6 \times \frac{5\,nA}{\mu m} \right\} \times 1$

$= 275\,mW$

Other half of transistors are off and contribute to the subthreshold leakage, I_{sub}.

$P_{sub} = I_{sub} \times V_{DD}$

$= \frac{1}{2} \times \left\{ 0.75 \times 10^6\,\mu m \times \frac{100\,nA}{\mu m} + 109.25 \times 10^6\,\mu m \times \frac{10\,nA}{\mu m} \right\} \times 1V$

$= (584\,mW)$

Hence $P_{static} = P_{sub} + P_{gate} = (859\,mW)$

Suppose the technology you are using to design a VLSI system has $\lambda = 80$ nm, a clock frequency of **5 MHz**, and a supply is **5 V**. The chip you are designing has **5 million transistors**, of which 1 million remain **active** at any given time. The gate and diffusion capacitances are 12 fF/μm and 5 fF/μm, respectively. The gate width is **20λ**. You also obtain the following power consumption data:

- Short circuit power = 0.5 W
- Leakage power = 0.01 W
- Subthreshold power = 0.02 W

The **acceptable TOTAL power consumption** of a chip is **3 W.**

(a)	**Find** the activity factor and load capacitance of the system described above.	**[2]**
(b)	**Calculate** the switching power consumption of the chip.	**[3]**
(c)	**Calculate** the dynamic and static power of the chip. Thereafter calculate the **TOTAL** power consumption.	**[3]**
(d)	Is the **TOTAL** power consumption within the acceptable range? If not, **find** the maximum clock frequency to keep the **TOTAL** power within the acceptable range?	**[4]**

Answer:

$V_{DD} = 5V$, $f_{sys} = 5MHz$, $\lambda = 80nm$.

a) $\alpha = \frac{1 \times 10^6}{5 \times 10^6} = (0.2)$

total capacitance at load for 5M transistors $= (5\times10^6) \times (20\lambda) \times (12+5)\ fF/\mu m$

$= (5\times10^6) \times (20 \times 0.08\ \mu m) \times \frac{17\ fF}{\mu m}$

$= (0.136\ \mu F)$

b) $P_{sw} = (\alpha f) C_T V_{DD}^2 = (0.2 \times 5 \times 10^6) \times (0.136 \times 10^{-6}) \times (5)^2$

$= (3.4)W$

c) $P_{dynamic} = P_{sw} + P_{short-ckt} = 3.4 + 0.5 = (3.9)\ W$

$P_{static} = P_{leakage} + P_{sub} = 0.03$

$\therefore$ total Power consumption $= 3.9 + 0.03 = (3.93)W$

d) since, $P_{total} = 3.93W \boxed{> 3W} \rightarrow$ we need $P_{total} \leq 3W$

$\Rightarrow \{(\alpha f' C_T V_{DD}^2) + 0.5\} + \{0.01 + 0.02\} \leq 3W$

$\Rightarrow f'$ = new clock freq $\leq 3.63MHz \rightarrow \therefore$ max freq $= 3.63MHz$

A chip contains 1 billion transistors made from a 0.7 V 120 nm technology with $\lambda = 60$ nm. Out of them, 100 million are logic gates, 200 million are never on, and the rest are memory arrays. Suppose that the logic gates operate at 1 GHz, the memory arrays operate at 3 GHz, and the clock frequency is 10 GHz. For all transistors, the gate and diffusion capacitance are 1.2 fF/um and 1.7 fF/um. The average width for logic gates and memory arrays are 4λ and 5λ, respectively. The following power data are given for the entire chip,

- Gate leakage power = 10 W
- Junction leakage power = 20 W
- Subthreshold power = 5 W
- Short circuit power of the logic gates = 60 W
- Short circuit power of the memory arrays = 70 W

The maximum acceptable power consumption of the chip is 1 kW.

(a)	**Find** the activity factor of the logic gates and the memory arrays.	**[2]**
(b)	**Find** the total switching power of each of the three kinds of transistors.	**[6]**
(c)	**Find** the static, dynamic, and total power of the entire chip.	**[3]**
(d)	Is the total power consumption within the acceptable range? If not, **find** the supply voltage that will keep it within range.	**[4]**

(a) $\boxed{\alpha_{logic} = \frac{1}{10} = 0.1 \;,\; \alpha_{memory} = \frac{3}{10} = 0.3}$

(b) $V_{DD} = 0.7V,\; f = 10GHz,\; \lambda = 60nm$

Logic gates $\Rightarrow N = 100 \times 10^6,\; W = 4\lambda = 0.24\mu m$

~~$P_{sw1} = N\alpha f C V_{DD}^2$~~

$C = (1.2+1.7)fF/\mu m \times 0.24\mu m = 0.696 \times 10^{-15} F$

$\boxed{P_{sw1} = N\alpha f C V_{DD}^2 = 34.1W}$

Inactive $\Rightarrow \boxed{P_{sw2} = 0W}$

Memory arrays $\Rightarrow N = 700 \times 10^6,\; W = 5\lambda = 0.3\mu m$

$C = (1.2+1.7)fF/\mu m \times 0.3\mu m = 0.87 \times 10^{-15} F$

$\boxed{P_{sw3} = N\alpha f C V_{DD}^2 = 895.23W}$

(c) $\boxed{P_{static} = 10+20+5 = 35W}$

$\boxed{P_{dynamic} = 34.1+0+895.23+60+70 = 1059.33W}$

$\boxed{P_{total} = 1094.33W}$

(d) $P_{total} > 1kW$

$P'_{total} = 1000W$

$\therefore P'_{sw} = 1000 - 10 - 20 - 5 - 60 - 70$

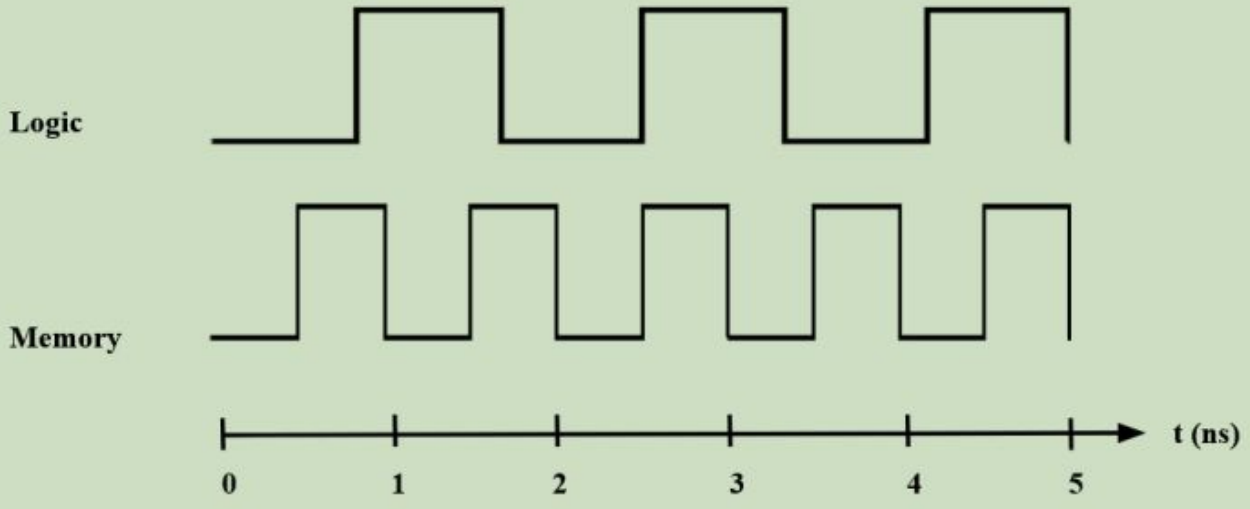
$= 835W$

$\left(\frac{P'_{sw}}{P_{sw}}\right) = \left(\frac{V'_{DD}}{V_{DD}}\right)^2$

$\boxed{\therefore V'_{DD} = V_{DD}\sqrt{\left(\frac{P'_{sw}}{P_{sw}}\right)} = 0.66V}$

$[P_{sw} = 34.1 + 895.23 = 929.33W]$

Suppose you are designing a chip in a 1.1 V, 45 nm technology which has 50 million transistors, of which 20 million are used for building logic gates and the rest are used in memory arrays. The average logic transistor width is 10λ and the average memory transistor width is 7λ. The chip has an area of 60 mm^2. The gate and diffusion capacitances are 15 pF/mm^2 and 80 pF/mm^2, respectively for a total 50 million transistors. Neglect the wire capacitance. The operating frequency is 5 GHz. The switching signals for both logic and memory transistors are shown in figure 4 (a).



You also obtain the following current data for each transistor:

- Gate leakage is 5 nA/μm.
- Junction leakage is 2 nA/μm
- Subthreshold leakage is 10 nA/μm

The acceptable total power consumption of a chip is 10 W.

(a)	**Find** the activity factor of the logic gates and the memory arrays.	**[2]**
(b)	**Calculate** the switching power consumption of the chip.	**[6]**
(c)	**Estimate** the static power of the chip. Thereafter, **calculate** the total power consumption.	**[3]**
(d)	Is the total power consumption within the acceptable range? If not, find the maximum clock frequency to keep the total power within the acceptable range.	**[4]**

a) T_{logic}= 5/3 ns; So, f_{logic}= 3/5 GHz= 0.6 GHz; So, α_{logic}= **0.6/5= 0.12**

T_{mem}= 5/5 ns; So, f_{mem}= 5/5 GHz= 1 GHz; So, α_{mem}= **1/5= 0.2**

b) Logic: 20mil, 10λ

$f = 5GHz$, $V_{DD} = 1.1V$, $\lambda = 45nm$, $A = 60\ mm^2$

Memory: 30mil, 7λ

$C_{per\text{-}mm} = 15 + 80\ pF/mm^2 = 95\ pF/mm^2$

$C_{total} = 60 \times 95\ pF$

$= 5.7\ nF$

$P_{switching} = (\alpha_{Logic} + \alpha_{mem})\, C_{total} \times f \times V_{DD}^2$

$= 11.0352\ W$

c) Logic: 10λ

$\lambda = 45nm$

$I_{single} = 17\ nA/um$

Memory: 7λ

$I_{L,total} = 20 \times 10^6 \times 17nA/um \times 16\lambda$

$= 0.153\ A$

$I_{M,total} = 30 \times 10^6 \times 17nA/um \times 7\lambda$

$= 0.161\ A$

$P_{static} = (I_{L,total} + I_{M,total})\, V_{DD}$

$= 0.3454\ W$

$P_{total} = P_{dyn} + P_{static}$

d) Not within range as $P_{total} = 11.3806W > 10W$

So, $P_{tot,max} = 10W$

$$P_{sw,new} = P_{tot,max} - P_{static}$$

$$= 10 - 0.3454$$

$$= 9.6546\,W$$

$$\frac{P_{sw,new}}{P_{sw,old}} = \frac{f_{new}}{f}$$

$$\Rightarrow \frac{9.6546}{11.3502} = \frac{f_{new}}{5\,GHz}$$

$$\therefore \boxed{f_{new} = 4.25\,GHz}$$